

## Problem-Solution Fit

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|----------------------|-------------------------------------|
| <b>Date</b>          | 15 July 2026                        |
| <b>Team ID</b>       | SWUID2026217449                     |
| <b>Project Name</b>  | Rising Waters - Flood Prediction ML |
| <b>Maximum Marks</b> | 5 Marks                             |

### Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

|                                                                                                                                    |                                                                                                                                                          |                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>1. CUSTOMER SEGMENT(S) [CS]</b><br>Meteorologists, disaster management teams, and local authorities in flood-prone regions      | <b>6. CUSTOMER LIMITATIONS</b><br>EG. BUDGET, DEVICES [CL]<br>Limited internet access in remote areas; budget constraints for advanced forecasting tools | <b>5. AVAILABLE SOLUTIONS</b><br>PROS & CONS [AS]<br>Manual weather analysis; government flood bulletins; basic rainfall gauges |
| <b>2. PROBLEMS / PAINS + ITS FREQUENCY [PR]</b><br>Frequent floods with inadequate advance warning, occurring every monsoon season | <b>9. PROBLEM ROOT / CAUSE [RC]</b><br>Weather data (rainfall, humidity, temperature) is not analyzed together using ML models                           | <b>7. BEHAVIOR + ITS INTENSITY [BE]</b><br>High anxiety and urgency to check conditions during heavy rainfall                   |
| <b>3. TRIGGERS TO ACT [TR]</b><br>Sudden heavy rainfall or rising humidity levels triggers the need to check flood risk            | <b>10. YOUR SOLUTION [SL]</b><br>Rising Waters ML-based flood prediction system<br>Rising Waters ML-based flood prediction system                        | <b>8. CHANNELS OF BEHAVIOR [CH]</b><br><br>ONLINE<br>Web browser access; word of mouth among disaster response teams            |
| <b>4. EMOTIONS BEFORE / AFTER [EM]</b><br>Before: anxious and uncertain / After: informed and prepared                             |                                                                                                                                                          | OFFLINE                                                                                                                         |